ROBOTICS · CHART 2

ESP32 car + ESP32-C3 gateway

MarkerPilot

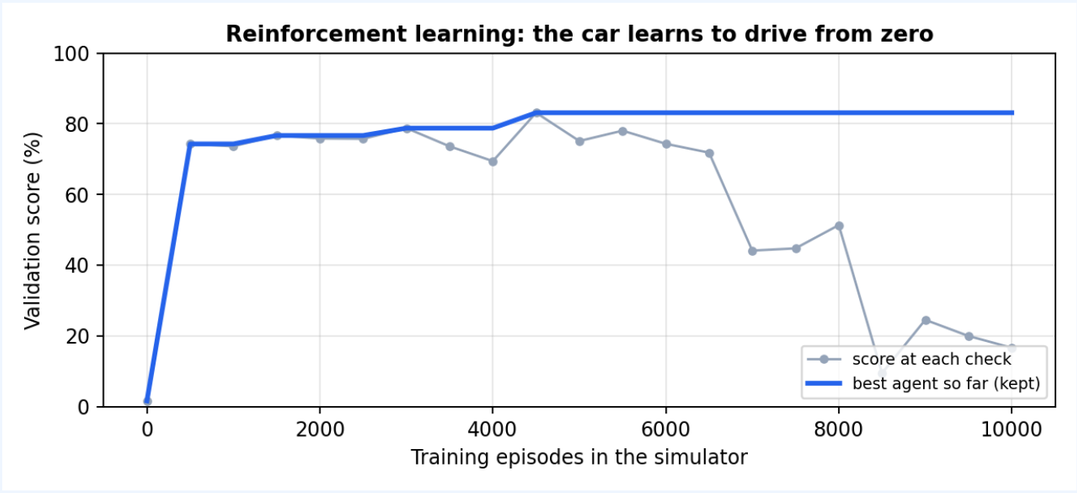
Every camera frame: see, filter, decide, stay safe, drive

30 decisions every second · ESP-NOW radio · Windows, Linux and macOS app

1 Every camera frame

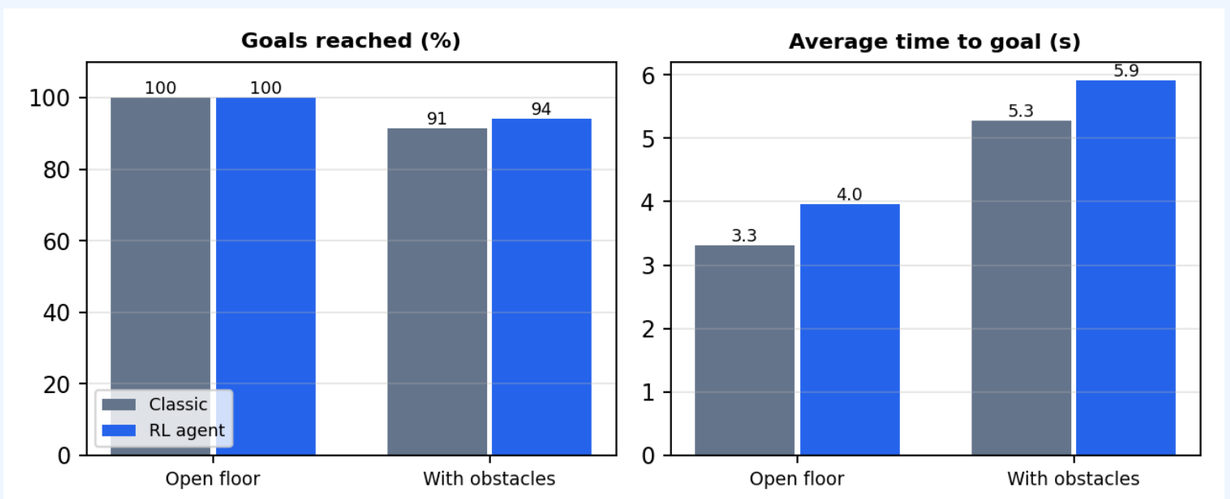
1. Grab a camera frame (30 / s)
2. Find the ArUco markers
3. Kalman filter: position, speed, heading
4. Pick target: click or goal marker
5. Brain: classic or RL agent
6. Safety: avoid, detour, escape
7. Send wheel speeds -> radio -> car

2 Learning to drive (RL)



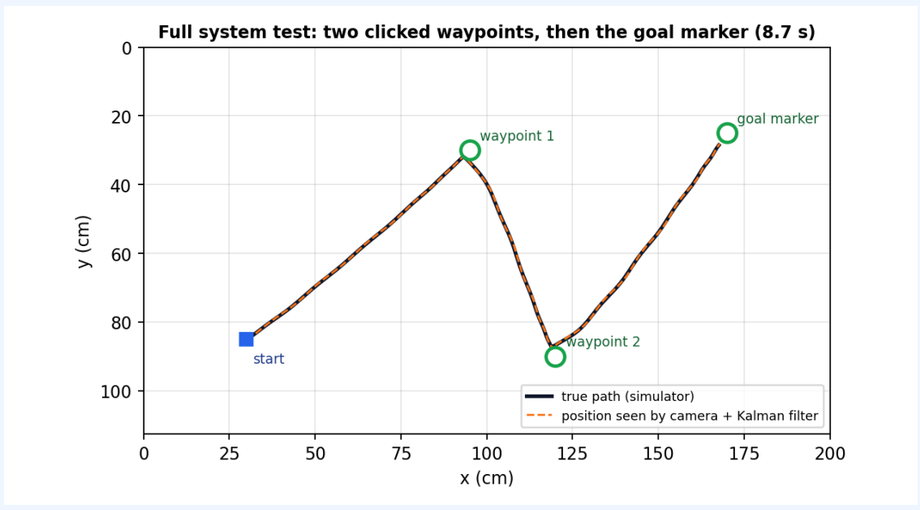
Starting from nothing, the agent learns to reach ~80 % of goals by practising in the simulator.

3 Classic controller vs RL agent



300 randomised simulator runs each: different motors, delays, goals and obstacles.

4 A full test run



Two clicked waypoints, then the goal marker. Dashed = what the camera believed.

5 Reliable radio (v2)

- Car and gateway **pair by themselves**.
- Motor power **ramps**: no brown-outs.
- Stops if silent for **400 ms**.
- On-car **collision guard**.
- **linktest**: signal, loss, delay.

6 Uses and next steps

Uses: warehouse and hospital delivery robots · factory floors · classroom AI kit. **Next:** camera obstacle detection with A* paths · several cameras · many cars at once.